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Original article

Evaluation of techniques used to assess skeletal muscle quantity in patients with cirrhosis

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SUMMARY

Introduction: Loss of skeletal muscle mass is a well-recognised complication of cirrhosis. Bedside methods to assess skeletal muscle mass including anthropometrics and bioelectrical impedance analysis (BIA) are negatively impacted by fluid overload in advanced cirrhosis and thus there is a need to identify alternatives. There is a paucity of data on the accuracy of commonly used radiological methods such as dual X-ray absorptiometry (DXA) to assess appendicular lean mass (ALM), and computed tomography (CT) skeletal muscle area in patients with cirrhosis. The aim of this study was to evaluate the relationships and agreement of several skeletal muscle mass estimation methods compared to a reference model in patients with cirrhosis.
Patients and methods: A cross-sectional, single centre study was performed by prospectively recruiting patients with cirrhosis referred to the Queensland Liver Transplant Service. Patients underwent assessment of skeletal muscle mass using bedside techniques (mid-upper arm muscle circumference (MUAMC), bioelectrical impedance spectroscopy (BIS), ultrasound muscle thickness (USMT)) and radiological methods (DXA ALM, CT skeletal muscle area). These were compared to a reference measurement of body cell mass derived from a multi-compartment model using isotope dilution tests and DXA.
Results: Forty-two patients (age 56 years, interquartile range 48–60, 86% male) were recruited. Bedside skeletal muscle mass estimation techniques were strongly correlated to the body cell mass reference, with BIS estimation having the strongest correlation coefficients ($r = 0.78–0.79$; $P < 0.01$). A novel technique measuring USMT offered no advantage over traditional bedside techniques. Of the radiological methods, DXA ALM had the strongest correlation coefficient ($r = 0.781$; $P < 0.01$). Weaker correlation coefficients were observed in patients with ascites, except when using the MUAMC. Bland-Altman analysis of BIS body composition estimates demonstrated significant systematic biases and large limits of agreement compared to reference values.
Conclusion: These results confirm the difficulties in assessing skeletal muscle mass in patients with cirrhosis, particularly in those with ascites. DXA ALM and BIS measurements provided the best correlation to body cell mass. We suggest DXA ALM for estimation of skeletal muscle mass in patients with cirrhosis as there are established thresholds for skeletal muscle mass depletion, and an accurate assessment of bone mass and density can also be provided. The use of USMT over other bedside skeletal muscle mass estimates was not supported by our results. Further studies evaluating novel bedside skeletal muscle mass estimation techniques in cirrhosis patients are required.
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Abbreviations Page

BCM	Body cell mass
BIA	Bioelectrical impedance analysis
BIS	Bioelectrical impedance spectroscopy
BMI	Body mass index
CT	Computed tomography
DXA	Dual energy X-ray absorptiometry
ECW	Extracellular water
ICC	Intraclass correlation coefficient
MUAMC	Mid-upper arm muscle circumference
SAT	Subcutaneous adipose tissue
SMA	Skeletal muscle area
SMI	Skeletal muscle index
TBK	Total body potassium
TBW	Total body water
US	Ultrasound
VAT	Visceral adipose tissue

1. Introduction

Loss of skeletal muscle mass is a well-recognised complication of cirrhosis which is associated with morbidity and mortality. In 2001, Alberino et al. used anthropometry (mid-upper arm muscle circumference [MUAMC]) to demonstrate that skeletal muscle mass depletion was an independent risk factor for mortality [1].

Since then, there have been multiple publications exploring the implications of skeletal muscle mass depletion in patients with cirrhosis. Many of these studies have used body composition techniques such as bioelectrical impedance analysis (BIA), which are based on a two-compartment model (fat-mass/fat-free mass) and rely on prediction equations to estimate skeletal muscle mass. However, patients with cirrhosis have alterations in the hydration fraction and density of fat-free mass even in early stages of disease, and prediction equations have been shown to be inaccurate in these patients [2,3]. With the introduction of multi-frequency systems, it was proposed that differentiation of these water compartments using bioelectrical impedance spectroscopy (BIS) was possible [4]. However, there may still be accuracy issues in patients with cirrhosis when compared to a gold standard four-compartment model (total body protein/total body water/bone mineral content/fat mass) [3].

More recently, cross-sectional imaging such as computed tomography (CT) has allowed the direct assessment of skeletal muscle mass and adipose tissue. In theory, this should counteract many of the inaccuracies associated with prediction equations. However, this is yet to be validated in patients with cirrhosis, with concerns being raised about the accuracy of this technique in patients with ascites due to potential disturbances in CT attenuation values [5]. Another radiological technique that offers direct assessment of skeletal muscle mass at the tissue level is ultrasound (US). Campbell et al. described a novel skeletal muscle mass estimation technique involving measuring the skeletal muscle thickness (USMT) at three appendicular sites (upper arm, forearm, and thigh) [6]. However, the accuracy of this method in patients with cirrhosis is also yet to be evaluated.

Prior studies in patients with cirrhosis that have evaluated the accuracy of body composition methods against a reference model have suggested that dual energy X-ray absorptiometry (DXA) is superior to most other two-compartment techniques [3], and bedside techniques for estimating fat mass and fat-free mass [2]. It has been suggested that DXA appendicular lean mass (ALM) is not affected by ascites and is therefore the preferred method to assess skeletal muscle mass [7]. However, DXA ALM has not been compared to a reference model in patients with cirrhosis.

The aim of this study was to evaluate the relationships and agreement of traditional and novel bedside skeletal muscle mass estimation techniques (MUAMC, BIS, USMT), and radiological techniques (DXA, CT) to reference measurements in patients with cirrhosis.

2. Materials and methods

This cross-sectional cohort study recruited patients with cirrhosis referred to the Queensland Liver Transplant Service between May 2014 and January 2017. Ethical approval was granted from the Metro South Hospital and Health Services District and the University of Queensland's Human Research Ethics Committees (Approval numbers: HREC/14/QPAH/23; 2014000254). Written informed consent was obtained prior to participation. Patients were excluded if they were less than 18 years of age and/or had previously received a liver transplant. Forty-two patients were included in the analysis.

Patients were assessed in the outpatient department following an overnight fast. Clinical data was collected including patient demographics, medical history, surgical history and prescribed medications. Biochemical results were recorded including bilirubin, albumin, international normalised ratio (INR), creatinine, and sodium.

2.1. Anthropometrics

Body weight and height were measured by a single observer (AW) to the nearest 0.1 kg and 0.1 cm using standardised equipment and procedures [8].

Mid-upper arm circumference (MAC) and triceps skinfold (TSF) measurements were taken on the right side at the mid-acromiale-radiale point. Three measurements were performed by an experienced observer (MW) with International Society for the Advancement of Kinanthropometry (ISAK) accreditation. The average value for all measurements was calculated for the analysis. The mid-upper arm muscle circumference (MUAMC) was calculated according to the following formula:

$$\text{MUAMC (cm)} = \text{MAC} - \pi\text{TSF}$$

The threshold used to define low skeletal muscle mass was <10th percentile of the sex and age-specific MUAMC [9].

2.2. Bioelectrical impedance spectroscopy

Right-side whole body bioelectrical impedance spectroscopy (BIS) was performed using a Bioimp SFB7 (Impedimed, Brisbane, Australia) as described previously [10].

Equations to estimate skeletal muscle mass in healthy populations using bioelectrical impedance data have been developed by several groups. These have not been evaluated in patients with cirrhosis. Skeletal muscle mass was predicted from the impedance data using the published equation from Tengvall et al. as follows [11].

BIS whole body skeletal muscle mass:

$$SMM_{Tengvall} = -24.05 + (0.365*Ht)(-0.005*Ri) + (-0.012*Re) + (-1.337*sex\ (M = 0, F = 1))$$

where Ri = intracellular resistance; Re = extracellular resistance.

Whole body skeletal muscle index (SMI) was calculated by adjusting for height, squared [12]. Low whole body SMI was defined as less than ≤8.50 kg/m^2 for men and ≤5.75 kg/m^2 for women [13].

2.3. Ultrasound muscle thickness

Ultrasound (US) estimation of skeletal muscle mass was performed using a Phillips EPIQ Ultrasound Scanner (Phillips, Andover, MA, USA) with a linear transducer 12-5 MHz by two blinded observers (AW and SK). The primary observer was an experienced sonographer (SK) while the other was an inexperienced observer (AW) who received basic sonography training. Three muscle thickness measurements were taken at the mid forearm, mid upper arm anteriorly and mid-thigh on the dominant side of the body as described previously [6]. Inter-observer variability was assessed on 10 patients with both observers performing a blinded examination on the same day. The intraclass correlation coefficient (ICC) was 0.97 (95%CI 0.87–0.99) demonstrating excellent agreement. Intra-observer variability was not assessed due to logistical constraints.

2.4. Computed tomography

Computed Tomography (CT) of the abdomen performed for clinical indications were available for reanalysis to assess skeletal muscle and adipose tissue. The time interval between the CT scan and other body composition assessments was recorded. Abdominal CT scans were assessed for quality as previously described [14]. Data were obtained from a single transverse CT image at the mid third lumbar vertebrae using ImageJ software (National Institute of Health, Bethesda, MD, USA). Tissue segmentation was performed using Hounsfield unit thresholds of −29 to +150 for muscle and −190 to −30 for adipose tissue [14,15]. Cross-sectional area of SAT, VAT and skeletal muscle was measured. Skeletal muscle area (SMA) was normalized for stature (cm [2]/m^2), and described as the skeletal muscle index (SMI) as previously reported [16].

Thresholds used to define low CT skeletal muscle index (Class I skeletal muscle mass depletion) were less than 48.9 cm [2]/m^2 for males and 36.3 cm [2]/m^2 for females [17]. These thresholds represent values one standard deviation from a 'young healthy' reference and are similar to the cut-offs used by Prado et al. which were shown to have prognostic significance in patients with cirrhosis [18].

All images were analysed by a single trained observer (AW). Intra-observer variability was assessed by re-analysing 10 randomly selected scans after a six-month interval. The ICCs suggested excellent agreement. The value for skeletal muscle area was 0.99 (95%CI 0.98–0.99), while for VAT this was 1.00 (95%CI 0.99–1.00) and for SAT 1.00 (95%CI 1.00–1.00). Coefficient of variation (CV) values were 2.0%, 1.7% and 0.9% respectively. To examine potential variation in image analysis software, the same scans were analysed using Slice-O-Matic (Tomovision, Montreal, QC, Canada), with ICCs of 0.99 (95%CI 0.95–0.99), 1.00 (95%CI 0.99–1.00) and 1.00 (95%CI 1.00–1.00) for skeletal muscle area, VAT and SAT, respectively. Inter-observer variability was determined by a radiologist (JR) blinded to results for AW, using ImageJ. The ICCs were consistent with excellent agreement. The values were 0.99 (95%CI 0.97–1.00), 1.00 (95%CI 0.99–1.00) and 0.99 (95%CI 0.99–1.00) for skeletal muscle area, VAT and SAT, respectively.

2.5. Dual-energy X-ray absorptiometry

A whole body DXA scan was performed using a GE Prodigy Scanner (GE Healthcare Lunar, Madison, WI, USA). The precision (CV) of the GE Prodigy has been estimated at 0.8% for lean tissue, 2.5% for fat mass, and 1.3% for bone mass in healthy cohorts [19]. DXA fat mass, fat-free mass, bone mass and appendicular lean mass (ALM) were measured according to manufacturer's guidelines by the duty radiologist. ALM was used to calculate the DXA Skeletal Muscle Index (SMI) according to the following equation: DXA SMI (kg/m^2) = DXA ALM/Height [2]. [20] Depletion of skeletal muscle mass was defined as a DXA SMI less than 7.26 kg/m^2 for men and 5.45 kg/m^2 for women [12,20].

2.6. Total body potassium

Total body potassium counting was performed using a shadow-shield whole-body counter (Accuscan, Canberra Industries, Boston, MA) containing three sodium iodide crystal scintillation detectors arranged over a scanning bed. Each subject underwent two 20-min measurements and the average was used for analysis. The CV for this machine is 7.1%.

Body cell mass was calculated using the following equation from Wang et al. [21]:

$$BCM_{TBK}\ (kg) = (TBK\ (g)\ x\ 9.20)/39.1$$

2.7. Body water assessment

Assessment of water compartments was performed using deuterium (D_2O) and bromide (Br) dilution tests. Venous blood was drawn immediately before the oral intake of a cocktail containing approximately 1g D_2O/kg body weight (using a solution of 99.9% D_2O) and 0.75 mmol Br/kg body weight. Venous blood samples were collected following a 4-h and 6-h equilibration period during which the patient remained fasting. Deuterium and bromide assays were performed using Fourier transform infra-red spectroscopy (Shimadzu Corp., Kyoto, Japan) and high performance liquid chromatography respectively [10]. Analytical precision using these methods has been estimated at <2% [22].

Total body water (TBW) was calculated from the quantity of deuterium administered and the measured increase in tracer concentration over the time course of the study, compared with baseline plasma concentration as described previously [22]. Extracellular water (ECW) volume was calculated as the corrected bromide space as described previously [22]. The bromide dilution space was converted into ECW by correcting for the fraction of water in plasma (0.94), the Donnan equilibrium factor for bromide (0.95), and the fraction of bromide dose that is retained and assumed to remain extracellular (0.90) [10].

Intracellular water (ICW) was calculated as the difference between TBW and ECW and the percentage hydration of fat-free mass was calculated as the ratio of TBW to DXA fat-free mass.

2.8. Body cell mass reference model

A multi-compartment 'reference' measurement of body cell mass was calculated as described by Wang et al. for comparison with skeletal muscle mass estimates [21]. Wang et al. described a body cell mass model as follows [21].

$$BCM_{ref}\ (kg) = FFM_{DXA}\ (kg) - \text{extracellular solids (ECS) (kg)} - \text{extracellular fluid (ECF) (kg)}$$

where ECS = 1.732 x DXA bone mass/0.9582 and ECF = ECW/*b* where *b* is the fraction of ECF that consists of ECW. *b* has been estimated at 0.98 [21].

3. Statistical analysis

Statistical analyses were performed using MedCalc Statistical Software (Version 17.9.7; Ostend, Belgium). Continuous data were assessed for normality using a Shapiro–Wilk test and are presented as means ± SD (for normally distributed data), or medians and IQR (for non-normally distributed data). Non-normally distributed data were transformed (logarithmic, square or exponential) where possible for statistical analysis.

Inter-observer and intra-observer variability were assessed using intraclass correlation coefficients. ICC values were interpreted as: 0.5–0.75 fair reproducibility, 0.75–0.9 good reproducibility, and >0.9 excellent reproducibility [23].

The relationship between skeletal muscle mass estimates and the body cell mass reference was assessed using Pearson's (r_p) correlation.

Due to logistical issues, data using all techniques was not available for all participants. CT data was excluded if the time between the two assessments was greater than three months, leaving 35 patients with data for inclusion. One patient did not have a pre-transplant CT available for analysis.

A sensitivity analysis was conducted using data from participants who underwent all measures (n = 25) and the correlation coefficients were similar to those presented [Supplementary Table 1].

The relationship between reference body composition values and estimates of fat mass, fat-free mass, TBW and ECW was assessed using Pearson's (r_p) and concordance correlation (r_c) as well as by limits of agreement analysis according to the method of Bland and Altman [24]. For interpretation of Pearson correlation coefficients Cohen's levels were used (r_p < 0.3 = weak; r_p 0.3–0.5 = moderate; r_p 0.5–0.8 = strong) [25]. Concordance correlation coefficients were interpreted using similar levels (r_p < 0.2 = weak; r_p 0.2–0.5 = moderate; r_p 0.5–0.8 = strong; r_p > 0.8 = excellent) as suggested by Altman [26].

4. Results

Forty-two patients (36 men, 6 women) with a median age of 56 years (interquartile range [IQR] 48–60) were included in the analysis. Patient clinical characteristics are shown in Table 1. The majority of the study cohort were male (86%) and Caucasian (88%). The most common causes of liver disease were hepatitis C (62%), alcohol (12%) and primary sclerosing cholangitis (7%). Twenty-two patients (52%) had a diagnosis of hepatocellular carcinoma (HCC).

Body composition results are shown in Table 2. The cohort had a mean weight of 82.5 ± 16.1 kg, with a median DXA fat mass of 21.0 kg (IQR 15.1–27.6) and mean DXA fat-free mass of 60.2 ± 10.5 kg. The mean reference body cell mass (BCM_{ref}) was 34.3 ± 7.2 kg. Bland and Altman analysis of body cell mass measured by total body potassium counting was compared to the reference body cell mass measurement ($BCM_{DXA/Br}$). Compared to the $BCM_{DXA/Br}$, the BCM_{TBK} had a significant mean bias of 14.9 kg (P < 0.001) with LOA ±11.4 kg.

4.1. Skeletal muscle mass estimation

Comparison of skeletal muscle mass estimation techniques with reference body cell mass models was performed using Pearson correlation [Table 3].

Traditional bedside measurements (MUAMC, BIS) correlated strongly with the reference body cell mass measurement. Upper limb ultrasound muscle thickness measurements also strongly correlated to the body cell mass reference, however thigh measurements had at best only moderate correlation (r_p = 0.464). Total US muscle thickness results had strong correlation to the reference methods. Isolating upper limb muscle thickness measurements (USMT arm) did not improve the correlation coefficients. CT skeletal muscle area and DXA SMI were both strongly correlated to body cell mass. BIS and DXA ALM measurements had the highest correlation coefficients compared to the body cell mass reference.

Table 1
Patient characteristics.

Characteristic	Result
Number of patients	42
Age	56 (48–60)
Sex: male [n (%)]	36 (86%)
Ethnicity [n (%)]	
Caucasian	37 (88%)
Asian	3 (7%)
Indigenous	1 (2%)
Other	1 (2%)
Aetiology of liver disease [n (%)]	
Chronic hepatitis C	26 (62%)
Alcohol	5 (12%)
Primary sclerosing cholangitis	3 (7%)
Other	8 (19%)
Hepatocellular carcinoma [n (%)]	22 (52%)
Laboratory values	
Bilirubin (μmol/L)	37 (22–61)
Albumin (g/L)	31 ± 6
INR	1.4 (1.2–1.6)
Creatinine (μmol/L)	69 (58–86)
Sodium (mmol/L)	136 ± 3
MELD score	14 ± 5
Child-Pugh class [n (%)]	
A	15 (36%)
B	15 (36%)
C	12 (29%)
Ascites [n (%)]	
None	21 (50%)
Mild/diuretic controlled	15 (36%)
Moderate/severe	6 (14%)

Values presented as mean (±SD), or median (interquartile range) where data were not normally distributed, and number and percentage. MELD - Model for end-stage liver disease; INR – International normalised ratio.
Laboratory reference ranges: Bilirubin <20 μmol/L; Albumin 35–50 g/L; INR 0.9–1.2; Creatinine 45–90 μmol/; Sodium 135–145 mmol/L.

4.2. Effect of ascites on skeletal muscle mass measurements

To evaluate the impact of overt ECW excess on skeletal muscle mass estimation, results were re-analysed after patients were divided according to presence or absence of ascites on clinical examination. There were 21 patients with and 21 without clinically detected ascites.

Comparison of skeletal muscle mass estimates to the body cell mass reference was performed using Pearson correlation coefficients and is shown in Table 4 and Table 5. All skeletal muscle mass estimates except the MUAMC had stronger correlation coefficients in patients without ascites.

In patients without ascites, traditional bedside skeletal muscle mass estimates had inferior correlation coefficients to DXA ALM.

Evaluation of skeletal muscle mass estimates in patients with ascites showed that DXA appendicular lean mass, CT skeletal muscle area, mid-upper arm muscle circumference and bioelectrical impedance were strongly correlated with the body cell mass reference. US muscle thickness measurements had inferior correlation coefficients, and in some instances, there was no significant relationship to the body cell mass reference.

Table 2
Body composition measurements.

Characteristic	Men N = 36	Women N = 6	Overall
Anthropometrics			
Height (m)	1.77 ± 0.07	1.64 ± 0.05	1.75 ± 0.08
Weight (kg)	84.7 ± 15.4	69.3 ± 14.4	82.5 ± 16.1
BMI (kg/m^2)	27.2 ± 4.7	25.6 ± 5.0	26.9 ± 4.7
MUAMC (cm)	26.3 ± 2.4	22.8 ± 3.9	30.2 ± 3.9
Ultrasound[a]			
US MT upper arm (cm)	2.87 ± 0.31	2.13 ± 0.47	2.72 ± 0.46
US MT forearm (cm)	3.57 (3.21–3.74)	2.90 (2.58–3.12)	3.52 (3.10–3.63)
US MT thigh (cm)	4.15 ± 0.64	3.66 ± 0.73	4.05 ± 0.67
US MT total (cm)	10.52 ± 0.73	8.60 ± 1.25	10.13 ± 1.15
Dual energy X-ray absorptiometry			
Fat mass (kg)	20.6 (13.6–27.4)	21.8 (18.8–29.8)	21.0 (15.1–27.6)
Fat-free mass (kg)	62.7 ± 8.7	45.5 ± 8.6	60.2 ± 10.5
Bone mass (kg)	3.09 ± 0.52	2.32 ± 0.33	2.98 ± 0.57
Appendicular lean mass (kg)	24.9 ± 4.0	17.7 ± 4.2	23.9 ± 4.8
Appendicular SMI (kg/m^2)	7.99 ± 1.13	6.52 ± 1.49	7.78 ± 1.28
Deuterium and bromide body water			
Total body water [D_2O] (L)	50.4 ± 8.0	38.0 ± 7.9	48.6 ± 9.0
Extracellular water [Br] (L)	20.9 ± 5.8	15.4 ± 5.9	20.1 ± 6.1
Intracellular water (L)	29.5 ± 5.9	22.6 ± 8.0	28.5 ± 6.6
Hydration of fat-free mass	79.6% (77.3–81.7)	84.1% (80.7–86.8)	80.1% (77.3–83.0)
Total body potassium counting			
TBK (g)	96.4 ± 14.9	70.4 ± 9.0	93.2 ± 16.7
Bioelectrical impedance spectroscopy			
Fat mass (kg)	18.0 (12.1–27.9)	21.8 (19.5–27.2)	18.2 (13.3–27.6)
Fat-free mass (kg)	66.0 ± 11.8	45.7 ± 9.9	62.5 ± 12.7
Whole body SMM (kg) [Tengvall] [11]	27.5 ± 3.2	17.1 ± 6.1	27.6 (23.9–29.3)
Total body water (L)	47.5 ± 8.3	34.9 ± 8.0	45.7 ± 9.3
Extracellular water (L)	21.7 ± 3.7	15.4 ± 3.6	20.8 ± 4.2
Computed tomography			
Skeletal muscle area (cm^2)	168.4 ± 27.9	117.4 ± 12.3	160.9 ± 31.8
VAT area (cm^2)	112.3 (75.6–191.7)	66.4 (49.6–88.0)	104.9 (69.0–160.0)
SAT area (cm^2)	151.7 (92.3–207.3)	185.7 (95.9–260.3)	152.6 (97.2–211.3)
Body cell mass estimates			
BCM_{ref} (kg)	35.8 ± 6.0	25.6 ± 8.5	34.3 ± 7.2
BCM_{TBK} (kg)	22.7 ± 3.5	16.6 ± 2.1	21.9 ± 3.9

Values presented as mean (±SD), or median (interquartile range) where data were not normally distributed, and number and percentage.
BMI - Body mass index; MUAMC – Mid-upper arm muscle circumference; US MT – Ultrasound muscle thickness; D_2O – Deuterium; Br – Bromide; TBK – Total body potassium; BCM - Body cell mass; SMA – Skeletal muscle area; SMI – Skeletal muscle index; SMM; Skeletal muscle mass; VAT - Visceral adipose tissue area; SAT - Subcutaneous adipose tissue area.
[a] USMT Male n = 23; Female n = 6.

4.3. Bioelectrical impedance spectroscopy fat-free mass estimation

Comparison of fat-free mass measured by DXA and BIS was performed [Fig. 1]. Pearson correlation showed a strong relationship ($r_p = 0.931$; $P < 0.001$), while concordance correlation coefficient ($r_c = 0.899$) was excellent with a minor deviation from the line of identity. Limits of agreement (LOA) analysis demonstrated a significant mean systematic bias of −2.2 kg ($P = 0.005$) with large limits of agreement (±9.5 kg). There was a statistically significant weak inverse correlation ($r_p = -0.192$; $P = 0.003$) between the difference in DXA-measured and BIS predicted fat-free mass and the mean fat-free mass. Thus, BIS overpredicted fat-free mass with increasing mean fat-free mass values. In subjects with moderate-severe ascites, BIS underpredicted fat-free mass in most cases.

4.4. Bioelectrical impedance spectroscopy fat mass estimation

Estimation of fat mass by BIS was compared to DXA fat mass using concordance correlation and LOA analysis [Fig. 2]. Pearson correlation demonstrated a strong relationship ($r_p = 0.865$; $P < 0.001$), while the concordance correlation coefficient ($r_c = 0.842$) was excellent with a minor deviation from the line of identity. However, LOA analysis showed a significant systematic bias of +2.1 kg ($P = 0.005$) with large LOA (±9.2 kg). There was a weak but significant positive correlation ($r_p = 0.212$; $P = 0.016$) between the difference in DXA-measured and BIS predicted fat mass and the mean fat mass of the two methods. Thus, BIS underpredicted fat mass as fat mass increased. In those subjects with moderate-severe ascites, BIS overpredicted fat mass in most cases.

4.5. Bioelectrical impedance spectroscopy body water estimation

Comparison of TBW measured by isotope (deuterium) dilution was compared to BIS predicted TBW [Fig. 3]. The Pearson correlation coefficient revealed a strong relationship ($r_p = 0.889$; $P < 0.001$), while the concordance correlation coefficient ($r_c = 0.832$) was again excellent with a minor deviation from the line of identity. LOA analysis found a significant systematic bias of 2.9 L ($P < 0.01$) with large limits of agreement of ±8.0 L. The presence of moderate-severe ascites was associated with an underestimation of TBW.

Analysis of ECW measurements showed a considerably lower Pearson correlation coefficient ($r_p = 0.631$) and concordance correlation coefficient ($r_c = 0.590$) with a large deviation from the line of identity [Fig. 4]. LOA analysis found a non-significant mean

Table 3
Comparison of skeletal muscle mass estimates to a body cell mass reference (n = 42; men = 36, women = 6).

	Men		Women		Overall	
	r_p	*P* Value	r_p	*P* Value	r_p	*P* Value
MUAMC (cm) [c]	0.442	**0.011**	0.898	**0.015**	0.665	**<0.001**
$BIS_{Tengvall}$ [11] (kg)	0.702	**<0.001**	0.713	0.112	0.782	**<0.001**
US MT biceps (cm)[b]	0.218	0.319	0.829	**0.042**	0.636	**<0.001**
US MT forearm (cm)[b,e]	0.299	0.165	0.467	0.350	0.555	**0.002**
US MT thigh (cm) [b]	0.355	0.096	0.438	0.385	0.464	**0.011**
US MT total (cm) [b]	0.500	**0.015**	0.600	0.208	0.737	**<0.001**
US MT arm[a] (cm)[b,e]	0.379	0.074	0.738	0.094	0.645	**<0.001**
DXA ALM (kg)	0.699	**<0.001**	0.714	0.111	0.781	**<0.001**
CT SMA (cm^2) [d]	0.595	**<0.001**	0.943	**0.005**	0.752	**<0.001**

BCM - Body cell mass; TBK – Total body potassium; DXA – Dual energy X-ray absorptiometry; MUAMC – Mid-upper arm muscle circumference; BIS – Bioelectrical impedance spectroscopy; US MT – Ultrasound muscle thickness; ALM – Appendicular lean mass; CT – Computed tomography; SMA – Skeletal muscle area.
[a] US MT Arm = US MT biceps + forearm.
[b] n = 29.
[c] n = 37.
[d] n = 30.
[e] Analysis performed with transformed data.

negative bias of 0.7 L ($P = 0.350$), but large limits of agreement (±9.3 L). BIS consistently underestimated ECW in patients with moderate-severe ascites.

4.6. Comparison of skeletal muscle mass techniques for diagnosing skeletal muscle mass depletion

A complete body composition data set was available in 30 patients and used to assess the diagnosis of sarcopenia per modality. Previously published thresholds for diagnosis of skeletal muscle mass depletion were used for MUAMC, BIS, DXA and CT [9,12,13,17]. A comparison of the prevalence of skeletal muscle mass depletion from each of the approaches is shown in Fig. 5. MUAMC, $BIS_{Tengvall}$ and CT SMI all diagnosed skeletal muscle mass depletion in 8 patients (27%), while 7 patients were classified as having low skeletal muscle mass using DXA SMI.

The diagnosis of individual patients with skeletal muscle mass depletion was compared between the reference method (DXA SMI) and other modalities. The MUAMC resulted in the misclassification of 5 patients, while CT SMI misclassified 7 patients, and $BIS_{Tengvall}$ misclassified 5 patients.

Table 4
Pearson correlation coefficients comparing skeletal muscle mass estimates to reference body cell mass in patients without ascites (n = 21; men = 18, women = 3).

	Men		Overall	
	r_p	*P* Value	r_p	*P* Value
MUAMC (cm) [b]	0.308	0.230	0.524	**0.018**
$BIS_{Tengvall}$ (kg)	0.806	**<0.001**	0.858	**<0.001**
US MT biceps (cm)[c]	0.580	0.061	0.808	**<0.001**
US MT forearm (cm) [c]	0.117	0.731	0.514	0.060
US MT thigh (cm) [c]	0.544	0.083	0.291	0.312
US MT total (cm) [c]	0.696	**0.017**	0.782	**0.001**
US MT arm[a] (cm) [c]	0.523	0.099	0.784	**0.001**
DXA ALM (kg)	0.828	**<0.001**	0.877	**<0.001**
CT SMA (cm^2) [d]	0.718	**<0.001**	0.835	**<0.001**

BCM - Body cell mass; TBK – Total body potassium; DXA – Dual energy X-ray absorptiometry; MUAMC – Mid-upper arm muscle circumference; BIS – Bioelectrical impedance spectroscopy; US MT – Ultrasound muscle thickness; ALM – Appendicular lean mass; CT – Computed tomography; SMA – Skeletal muscle area.
Note: insufficient numbers for isolated analysis of female patients.
[a] US MT Arm = US MT biceps + forearm.
[b] n = 20.
[c] n = 14 (Men = 11; Women = 3).
[d] n = 18.

5. Discussion

The aim of this study was to evaluate the relationships and agreement of traditional and novel bedside skeletal muscle mass estimation techniques (MUAMC, BIS, USMT), and radiological techniques (DXA, CT) to reference measurements in patients with cirrhosis. When compared to a body cell mass reference, the BIS, mid-upper arm muscle circumference, DXA and CT skeletal muscle mass estimates all had strong and statistically significant correlation coefficients. For all measures, except the mid-upper arm muscle circumference, the relationships were weaker in patients with ascites. This highlights the accuracy issues of two-compartment body composition models in patients with cirrhosis, and is consistent with several prior publications [2,3].

US muscle thickness has been proposed as a novel technique to estimate skeletal muscle mass in patients with fluid retention, however our results do not support its use over BIS or the MUAMC. US muscle thickness had inferior correlation coefficients to the body cell mass reference, particularly in patients with ascites. Further refinement in technology and technique are required to improve its ability to estimate skeletal muscle mass before use in the clinical setting.

Despite the overall strong correlation observed with MUAMC and BIS assessments of skeletal muscle mass, these techniques have well documented limitations. MUAMC measurement has known issues with intra- and inter-observer variability, and requires specialised training and equipment [27]. BIS is associated with less observer variability, but also requires specialised equipment, and current prediction equations still assume a normal hydration fraction of fat-free mass [11,28,29]. Our results demonstrated there was a significant systematic bias in BIS measurements of TBW, fat-free mass and fat mass compared to reference measurements. These errors were more pronounced in patients with ascites and with increasing body mass. If bedside methods are used for the estimation of skeletal muscle mass, results should be interpreted with caution, particularly for patients with overt fluid overload.

Another area of concern with bedside skeletal muscle mass measurements was the discrepancy in diagnosis of skeletal muscle mass depletion using standard thresholds. Whilst the different modalities

Table 5
Pearson correlation coefficients comparing skeletal muscle mass estimates to reference body cell mass in patients with ascites (n = 21; men = 18, women = 3).

	Men		Overall	
	r_p	*P* Value	r_p	*P* Value
MUAMC (cm) [b]	0.614	**0.015**	0.802	**<0.001**
$BIS_{Tengvall}$ (kg)	0.657	**0.003**	0.765	**<0.001**
US MT biceps (cm)[c]	−0.199	0.536	0.465	0.095
US MT forearm (cm)[c,e]	0.426	0.167	0.604	**0.017**
US MT thigh (cm) [c]	0.217	**0.497**	0.618	**0.014**
US MT total (cm)[c,e]	0.282	0.123	0.697	**0.004**
US MT arm[a] (cm)[c,e]	0.156	0.628	0.481	0.070
DXA ALM (kg)	0.571	**0.013**	0.694	**0.001**
CT SMA (cm^2)[d,e]	0.486	**0.048**	0.649	**0.002**

BCM - Body cell mass; TBK – Total body potassium; DXA – Dual energy X-ray absorptiometry; MUAMC – Mid-upper arm muscle circumference; BIS – Bioelectrical impedance spectroscopy; US MT – Ultrasound muscle thickness; ALM – Appendicular lean mass; CT – Computed tomography; SMA – Skeletal muscle area.
Note: insufficient numbers for isolated analysis of female patients.
[a] US MT Arm = US MT biceps + forearm.
[b] n = 18.
[c] n = 15 (men = 12; women = 3).
[d] n = 20.
[e] Analysis performed with transformed data.

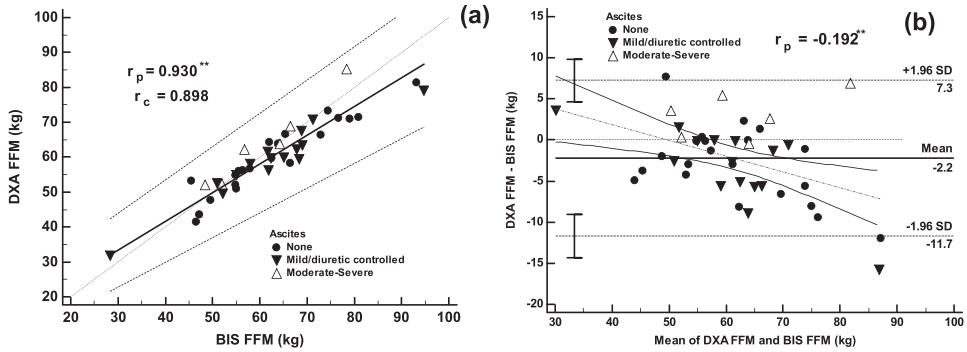

Fig. 1. Comparison of total body fat-free mass measured by dual energy X-ray absorptiometry (DXA) and bioelectrical impedance spectroscopy (BIS). (a) Scatter plot of DXA and BIS fat-free mass. Solid line - regression line; dashed line - 95% confidence interval; dotted line - line of identity. ** Significant at 0.01 level. (b) Limits of agreement analysis for total body fat mass measured by DXA and BIS. Dash-dot line - regression line with 95% confidence intervals; solid line - mean bias; dashed line - limits of agreement; dotted line - zero reference. ** Significant at 0.01 level.

resulted in a similar frequency of skeletal muscle mass depletion diagnosis, up to 8 patients per modality were misdiagnosed. This highlights the difficulty in comparing sarcopenia studies given the heterogeneity in skeletal muscle mass assessment modalities and thresholds used.

In this study, the relationship of DXA appendicular lean mass and CT skeletal muscle area to a body cell mass reference was also examined. Both DXA appendicular lean mass and CT skeletal muscle area have been advocated as accurate measures of skeletal muscle mass in the cirrhosis population [7,30]. DXA appendicular lean mass provided the strongest overall correlation coefficients to the reference body cell mass measurement. However, these coefficients are much lower than those observed when DXA appendicular lean mass was compared to whole body skeletal muscle mass in a healthy population ($r_p = 0.950$; $P < 0.01$) [31].

CT skeletal muscle area on a single abdominal slice has also been validated as an accurate predictor of total skeletal muscle mass in a healthy population [32]. However, some authors have expressed

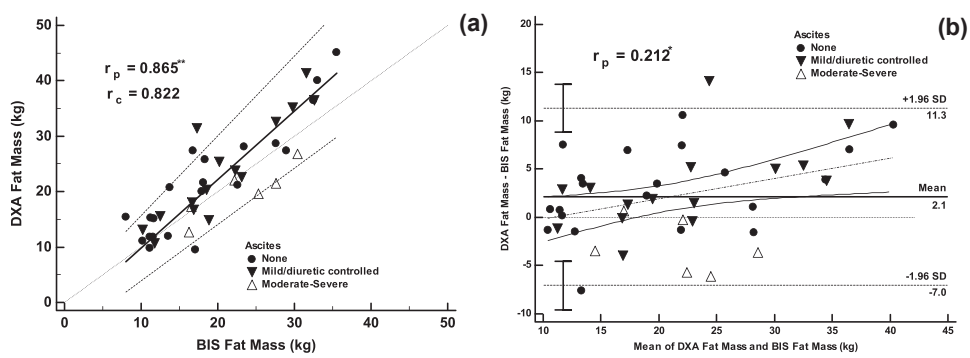

Fig. 2. Comparison of total body fat mass measured by dual energy X-ray absorptiometry (DXA) and bioelectrical impedance spectroscopy (BIS). (a) Scatter plot of DXA and BIS fat mass. Solid line - regression line; dashed line - 95% confidence interval; dotted line - line of identity. ** Significant at 0.01 level. (b) Limits of agreement analysis for total body fat mass measured by DXA and BIS. Dash-dot line - regression line with 95% confidence intervals; solid line - mean bias; dashed line - limits of agreement; dotted line - zero reference. * Significant at 0.05 level.

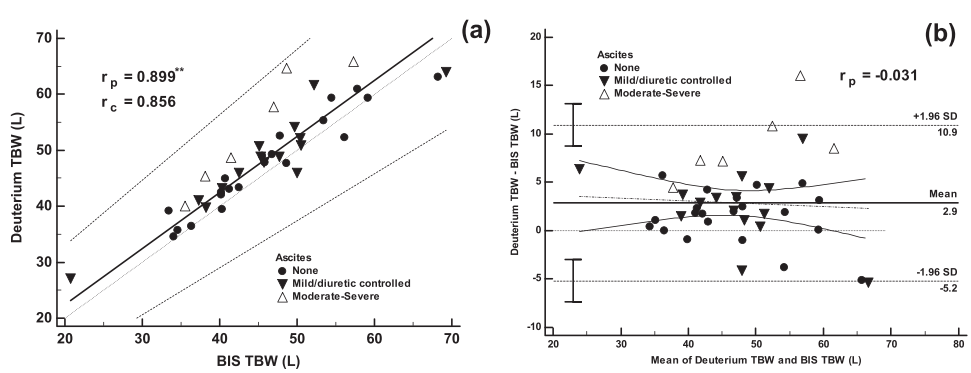

Fig. 3. Comparison of total body water (TBW) measured by deuterium and bioelectrical impedance spectroscopy (BIS). (a) Scatter plot of deuterium and BIS total body water. Solid line - regression line; dashed line - 95% confidence interval; dotted line - line of identity. ** Significant at 0.01 level. (b) Limits of agreement analysis for total body water measured by deuterium and BIS. Dash-dot line - regression line with 95% confidence intervals; solid line - mean bias; dashed line - limits of agreement; dotted line - zero reference.

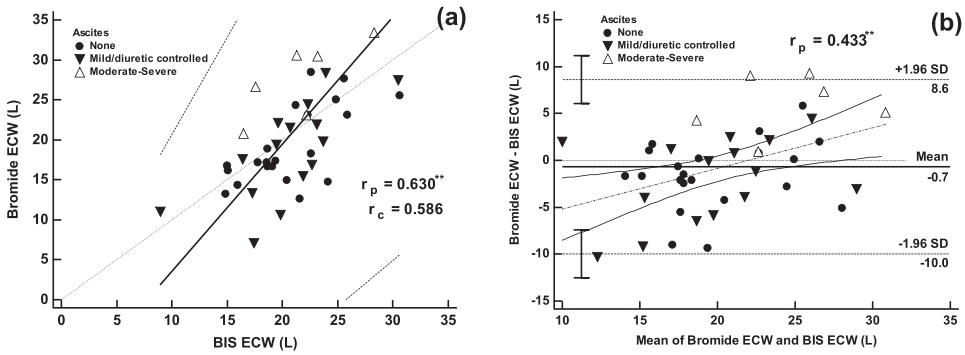

Fig. 4. Comparison of extracellular water (ECW) measured by bromide dilution and bioelectrical impedance spectroscopy (BIS). (a) Scatter plot of bromide and BIS ECW. Solid line - regression line; dashed line - 95% confidence interval; dotted line - line of identity. ** Significant at 0.01 level. (b) Limits of agreement analysis for bromide and BIS ECW. Dash-dot line - regression line with 95% confidence intervals; solid line - mean bias; dashed line - limits of agreement; dotted line - zero reference. ** Significant at 0.01 level.

concern regarding a lack of validation of CT estimation in patients with ascites [5]. Our results in patients with ascites supports this concern. The strength of correlation to the body cell mass reference declined in patients with ascites, and in male patients, there was no significant relationship detected. One explanation for the poor relationship is that muscle, ascites and oedema can have similar radiodensities and therefore cause interference with the image analysis software. Despite these concerns, abdominal CT measurement of skeletal muscle area has consistently demonstrated clinical significance in patients with cirrhosis and thus its value cannot be disregarded [33,34].

This is the first study, to the author's knowledge, to compare CT skeletal muscle area or DXA appendicular lean mass to a multi-compartment body cell reference method in patients with cirrhosis. However, it is important to note that this model, like most body composition models, has not been validated in patients with cirrhosis. With the multi-compartment model chosen, we believe the body cell mass measurement provides a suitable estimate of skeletal muscle mass, which takes into account body water compartments. Lack of a reference method which provided a direct measurement of total body protein or whole-body skeletal muscle mass (e.g. in-vivo neutron activation analysis and MRI respectively) is a limitation of this study. However, the use of tracer dilution tests to measure body water compartments and DXA to measure fat mass, fat-free mass and bone mass were considered a suitable alternative in patients with cirrhosis [2,3]. This model relies on the assumption that DXA is an accurate method of measuring fat mass and fat-free mass in patients with cirrhosis, which is supported by the study from Strauss et al. [3].

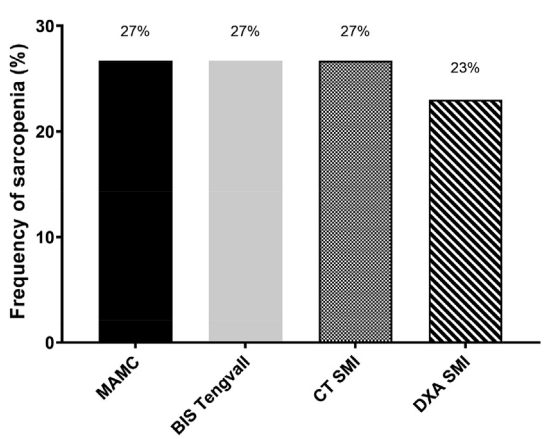

Fig. 5. Frequency of skeletal muscle mass depletion diagnosis according to modality (n = 30).

Total body potassium counting was utilised in our study and has been considered by some centres as a reference method for assessment of body cell mass and skeletal muscle mass, even in patients with excess ECW [35]. Other authors have expressed concern regarding the accuracy of TBK counting for body composition evaluation in patients with cirrhosis due to disturbances in total body potassium and intracellular potassium concentration [36,37]. The TBK estimates of body cell mass in our study were significantly lower ($P < 0.001$) than the reference body cell mass measurement. This is consistent with Strauss and colleagues who demonstrated a significant fat-free mass underestimation with TBK counting of 4.3 kg with wide LOA (±11.7 kg) [3].

We evaluated the utility of the latest BIS hardware and resistivity coefficients to assess fat mass, fat-free mass, TBW and ECW in patients with cirrhosis. Compared to reference measurements, the BIS results showed stronger correlation coefficients for fat mass, fat-free mass and TBW than the skeletal muscle mass measurements, although the limits of agreement remained large for all variables. The largest source of error appeared to be the BIS estimates of ECW, with a concordance correlation coefficient of 0.586 and large limits of agreement of ±9.3 L. BIS increasingly underestimated ECW as this increased. Therefore, BIS currently has limited utility in patients with increased ECW, as found in decompensated cirrhosis. However, technology advancements may be able to improve estimation of body water compartments, and allow the development of prediction equations, which account for disturbances in hydration status.

We suggest the use of DXA for evaluation of skeletal muscle mass in research and clinical practice where feasible and evaluation for low skeletal muscle mass using the thresholds of Baumgartner et al. [12] In our study, DXA appendicular lean mass had one of the strongest relationships to the body cell mass reference model. In addition, prior studies have demonstrated that is has excellent reproducibility in healthy adults [19,38]. DXA also provides an accurate assessment of bone mass and density, even in patients with a disturbance in hydration status [39]. However, use of DXA lean mass does have limitations in patients with cirrhosis, in particular patients with increased ECW, due its assumptions regarding the hydration of fat-free mass [2,3]. Use of the appendicular lean mass limits the impact of ascites on skeletal muscle mass estimation and has established depletion thresholds which are associated with adverse outcomes [7]. There is also the potential to use DXA in combination with BIS for estimation of total body protein using the model from Wilson et al. [40] However,

this model requires validation studies in patient with cirrhosis prior to routine clinical use.

Whilst bedside assessments including BIA, BIS and the MUAMC had strong relationships to the reference body cell mass, they have well known limitations in patients with cirrhosis. US muscle thickness measurements had inferior correlation coefficients and our data does not support their use in patients with cirrhosis. CT assessment of skeletal muscle mass is a reasonable alternative to DXA in patients who have had imaging for an alternative indication. Whilst our study supports prior concerns regarding accuracy in patients with ascites, abdominal CT muscle assessment has consistently demonstrated prognostic significance in patient with cirrhosis [33,34]. Further research is required to explore the impact of ascites on CT skeletal muscle estimation and whether refinement of body composition analysis software can improve the accuracy of CT. If prognosis is directly linked to whole body skeletal muscle mass, this would strengthen the prognostic value of CT SMI in patients with ascites. However, it is also feasible that CT SMI is a superior prognostic variable to whole body or appendicular skeletal mass. This could relate to an unmeasured confounding variable associated with abdominal CT SMI assessment.

Another important aspect of body composition analysis which was not evaluated in our study was functional body composition including resting energy expenditure and insulin resistance. As highlighted by Muller and colleagues, these are important variables in the study of body composition and may impact long term outcomes [41]. Further studies are required to explore these variables in patients with cirrhosis including their association with adverse outcomes.

To conclude, these results confirm the difficulties in assessing skeletal muscle mass in patients with cirrhosis, in particular those with significant fluid overload. Further refinement of established techniques, or development of novel bedside techniques is required to measure skeletal muscle mass in patients with cirrhosis.

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Author contributions

AW, MW, LW, JC, and GM were responsible for the study design; AW and MW were responsible for the data collection; JR contributed with radiological expertise and image analysis; AW, LW, JC, and GM contributed to the analysis and interpretation of data. AW, MW, LW, JC and GM contributed to the manuscript preparation and revisions.

Declaration of competing interest

AW, MW, JR, and GM have no personal or funding interests to disclose. LCW consults to ImpediMed Ltd. JC has received research funding from Renew Corp for unrelated work.

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Appendix A.Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.clnesp.2021.05.029.

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